

AI Security Lab — Level 02: AI Lies With Confidence

Vulnerability Overview

- **Vulnerability Type:** LLM Hallucination / Confidence Bias
- **Objective:** Demonstrate how an LLM can produce incorrect information in a fluent and confident manner and evaluate mechanisms for identifying and correcting the resulting misinformation.

Technical Analysis

Large Language Models can generate fluent and highly assertive responses even when the underlying information is incomplete, incorrect, or unsupported.

The apparent confidence of an LLM response should not be interpreted as evidence of factual correctness. Language generation is fundamentally probabilistic, and a well-formed response can still contain fabricated or inaccurate information.

This laboratory demonstrates the security and reliability implications of **LLM hallucination** and the risk of treating model-generated content as authoritative without independent verification.

Security Impact

Potential impacts include:

- Dissemination of inaccurate information.
- Incorrect automated decision-making.
- Misleading security or operational recommendations.
- Loss of data integrity and user trust.
- Increased risk when model outputs are consumed by downstream systems without verification.

Remediation

- Integrate **Retrieval-Augmented Generation (RAG)** where authoritative external knowledge is required.
- Implement **ground-truth verification** for security-sensitive or high-impact responses.
- Use trusted data sources and establish source attribution where appropriate.
- Introduce confidence and uncertainty handling rather than treating every model response as authoritative.
- Require human or automated verification for high-impact decisions.

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